

Complex Upper-Mantle Anisotropy Beneath the Pacific Triangle Revealed by Shear-Wave Splitting

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Abstract

Understanding mantle flow beneath old oceanic plates is essential for linking present-day plate motions with long-term lithospheric evolution, yet direct constraints remain sparse due to limited offshore observations. Here, we investigate upper-mantle seismic anisotropy beneath the Pacific Triangle, one of the oldest preserved fragments of oceanic lithosphere, using shear-wave splitting measurements from the Oldest-1 broadband ocean-bottom seismometer array. We obtained 128 high-quality measurements using teleseismic *XKS* and direct *S* phases as well as *ScS* phases to characterize upper-mantle anisotropy. Fast polarization directions (FPDs) show pronounced regional variability, with predominantly northwest–southeast to north–northwest–south–southeast orientations in the northwest and east–northeast–west–southwest orientations in the southeast, with a mean delay time of ~ 1.5 s, higher than the global average of 1.0 s. These FPD variations cannot be explained solely by absolute plate motion (APM) or paleospreading directions, and instead show systematic spatial correspondence with *P*-wave velocity anomalies imaged at depths of ~ 95 to 305 km. In the northern and western parts of the study region, FPDs exhibit more northward orientations relative to APM predictions. In the central-to-eastern region, FPDs exhibit systematic azimuthal variations around a high-velocity lithospheric domain, suggesting coupling between mantle deformation and a mechanically strong lithosphere. In addition, null measurements are observed near the Magellan Seamount Trail and the paleotriple junction, spatially coincident with low *P*-wave velocity anomalies. Together, these observations document laterally variable and depth-dependent anisotropy beneath the Pacific Triangle and provide new offshore constraints on upper-mantle deformation beneath the oldest Pacific seafloor.

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Supplemental Material

Introduction

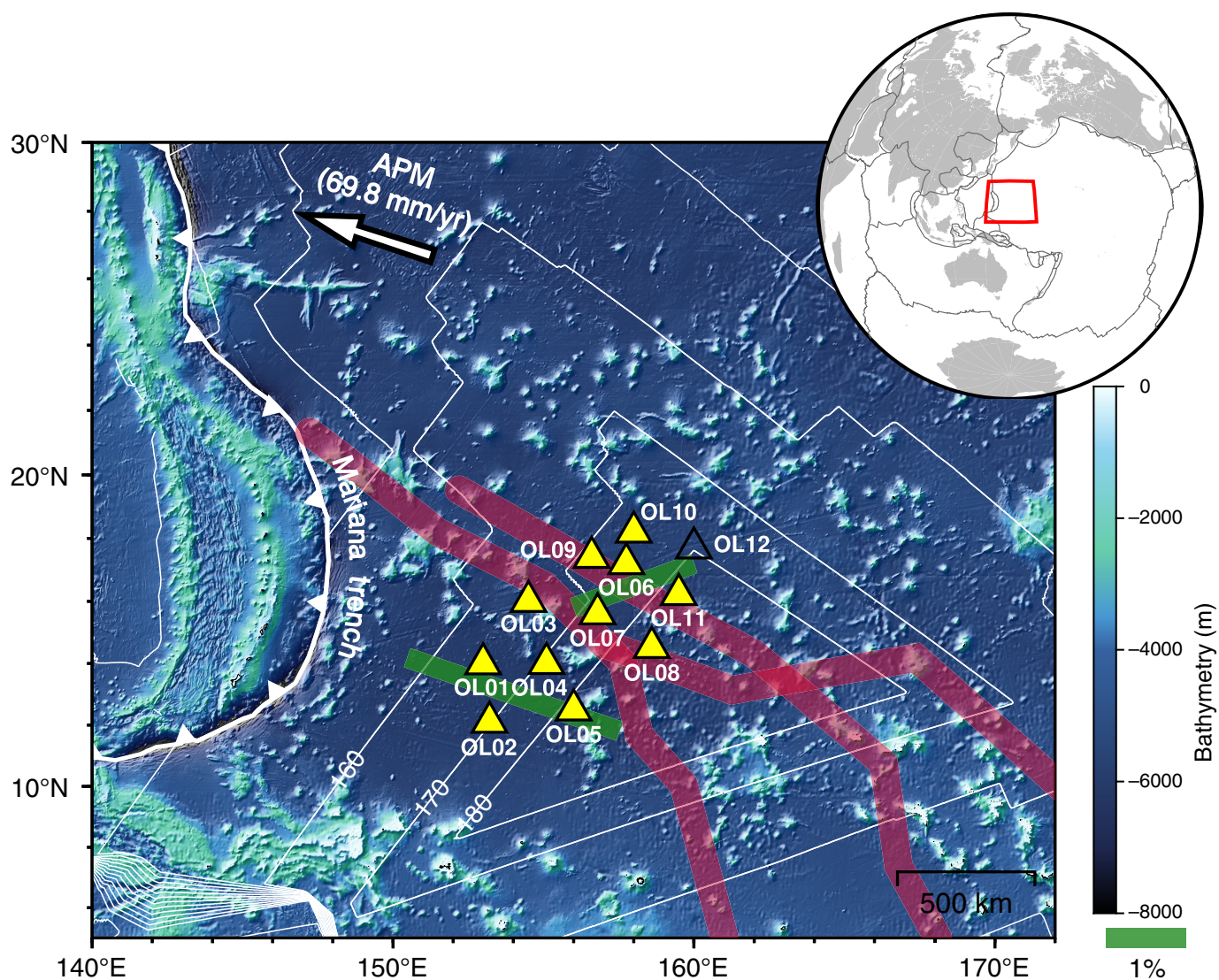
The flow of the Earth's uppermost mantle is a central topic in solid Earth sciences, as it controls plate motions and transports heat and materials, directly influencing the planet's thermal evolution. Therefore, understanding the geometry and driving forces of this flow is fundamental for linking plate tectonics with mantle dynamics. Such mantle flow aligns the crystalline structure of upper-mantle minerals, producing lattice-preferred orientations (LPOs) that can be detected through seismic anisotropy (Nicolas and Christensen, 1987; Karato, 2008; Skemer and Hansen, 2016). LPO developed in the asthenosphere primarily reflects present-day mantle flow, whereas anisotropy frozen into the lithosphere can persist as fossil anisotropy that records past mantle deformation (Nicolas and Christensen, 1987; Long and Silver, 2009). Disentangling these active and fossil components of seismic

anisotropy remains a long-standing challenge in seismology, particularly beneath old oceanic plates. Whereas young oceanic lithosphere is generally expected to show anisotropy aligned with present-day plate motion, old oceanic plates are more

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likely to host vertically stratified anisotropy, with fossil lithospheric fabrics and deeper asthenosphere flow-related anisotropy contributing at different depths (Smith *et al.*, 2004; Debayle and Ricard, 2013; Lynner and Long, 2014b).

Over the past decades, global seismic and geodynamic studies have sought to unravel the geometry and driving forces of mantle flow (e.g., Long and Silver, 2008; Becker *et al.*, 2014). Although the absolute plate motion (APM) is often invoked as the dominant control (Kreemer, 2009), numerous studies demonstrate that regional mantle flow can deviate from simple plate-driven shear (e.g., Lin *et al.*, 2016; Takeo *et al.*, 2018). Some processes have been proposed to account for such deviations. Mantle plumes may produce localized upwellings that disturb the background flow patterns (Ito *et al.*, 2014), whereas thick lithosphere or chemically depleted mantle roots can act as rigid keels, diverting surrounding asthenosphere flow (Fouch *et al.*, 2000; Li *et al.*, 2020). Such processes are expected to play an important role in the western Pacific, where the Pacific plate preserves the earliest record of oceanic spreading.

Figure 1. Bathymetric map of the study region showing the locations of the Oldest-1 array on the 160–180 Ma Pacific seafloor. Yellow filled triangles mark broadband ocean-bottom seismometer (BBOBS) stations equipped with differential pressure gauges (DPGs), and the open triangle marks OL12 where only DPG data were recorded. The white arrow indicates the absolute plate motion (APM) in the study region, based on Beavan *et al.* (2002) and expressed in the ITRF2000 reference frame (Altamimi *et al.*, 2002). Red shaded bands trace multiple hotspot tracks compiled by Jackson *et al.* (2024). Green bars show average fast polarization directions (FPDs) at depths shallower than 50 km derived from surface-wave dispersion analysis (Kawano *et al.*, 2023). White contours denote seafloor age (in Ma), and the inset map shows the regional location of the study area. The color version of this figure is available only in the electronic edition.

The Pacific Triangle, located ~1000 km east of the Mariana trench, represents the site where the Izanagi, Farallon, and Phoenix plates once converged at a ridge–ridge–ridge triple junction (Fig. 1). This region constitutes one of the oldest preserved fragments of oceanic lithosphere, and therefore

provides a rare opportunity to probe mantle deformation processes operating beneath mature oceanic plates. Magnetic lineations in this region diverge from present-day APM (Beavan *et al.*, 2002), and plate reconstruction models suggest that the absolute motion of the Pacific plate changed through time, potentially imposing different shear directions on the lithosphere and producing fossil fabrics distinct from present-day asthenosphere flow (Wessel and Kroenke, 2008; Boschman and van Hinsbergen, 2016; Wright *et al.*, 2016). These observations suggest that the Pacific Triangle may host complex, depth-dependent anisotropy resulting from the superposition of fossil lithospheric fabrics and ongoing asthenosphere flow, making it an ideal natural laboratory for testing competing models of mantle deformation.

Our study area lies within this Pacific Triangle and encompasses the East Mariana and Pigafetta basins, where seafloor ages of ~160 to 180 Ma represent the oldest preserved oceanic lithosphere on Earth (Müller *et al.*, 2008; Seton *et al.*, 2020; Fig. 1). The Magellan Seamount Trail (MST) transects the region and separates the East Mariana basin from the Pigafetta basin; seismic surveys along the trail indicate crustal thicknesses of ~8 to 10 km (Abrams *et al.*, 1993). Despite its tectonic significance, the structure and dynamics of the upper mantle beneath this region remain poorly constrained, largely due to the lack of dense seismic observations in this remote part of the Pacific.

To address this long-standing observational gap, a Korean–Japanese collaboration conducted the Oldest-1 experiment during 2018–2019, deploying 12 broadband ocean-bottom seismometers (BBOBSs) and seven ocean-bottom electromagnetometers (OBEMs) across the Pacific Triangle. The Oldest-1 array provides one of the first broadband seismic datasets capable of resolving mantle structure beneath the oldest Pacific seafloor. Previous surface-wave analyses of these data revealed a pronounced low-velocity zone at depths of ~80 to 240 km and significant lateral variations in azimuthal anisotropy (Kawano *et al.*, 2023). In addition, finite-frequency *P*-wave tomography imaged alternating high- and low-velocity anomalies at ~95 to 185 km depth, interpreted as signatures of upper-mantle heterogeneity potentially associated with plume residue or small-scale convection (SSC; Kang *et al.*, 2023). Together, these results point to a complex interplay between fossil lithospheric anisotropy and anisotropy associated with present-day asthenosphere flow beneath the Pacific Triangle, motivating further investigation using complementary seismic observables.

In this study, we explore upper-mantle seismic anisotropy using shear-wave splitting (SWS), a direct and widely used method for characterizing mantle deformation beneath seismic stations. When an *S* wave propagates through an anisotropic medium, it splits into fast and slow components whose polarization direction and delay time provide first-order constraints on the orientation and strength of anisotropy. As a station-

based observable, SWS provides a straightforward and intuitive measure of the anisotropic signature beneath each site, making it particularly effective for detecting lateral variations in mantle deformation beneath old oceanic plates.

Here, we present SWS measurements from BBOBS data recorded during the Oldest-1 experiment. Our results provide new observational constraints on upper-mantle anisotropy beneath the Pacific Triangle, where direct observations have been scarce. We document pronounced spatial variability in fast polarization directions (FPDs) and delay times, and evaluate these observations in the context of mantle flow potentially influenced by lithospheric heterogeneity, depth-dependent anisotropy, and regional mantle flow components oblique to APM. These findings offer new insights into how present-day mantle flow may be superimposed on pre-existing lithospheric fabrics beneath the oldest oceanic lithosphere, with broader implications for understanding mantle dynamics beneath long-lived oceanic plates.

Data and Methods

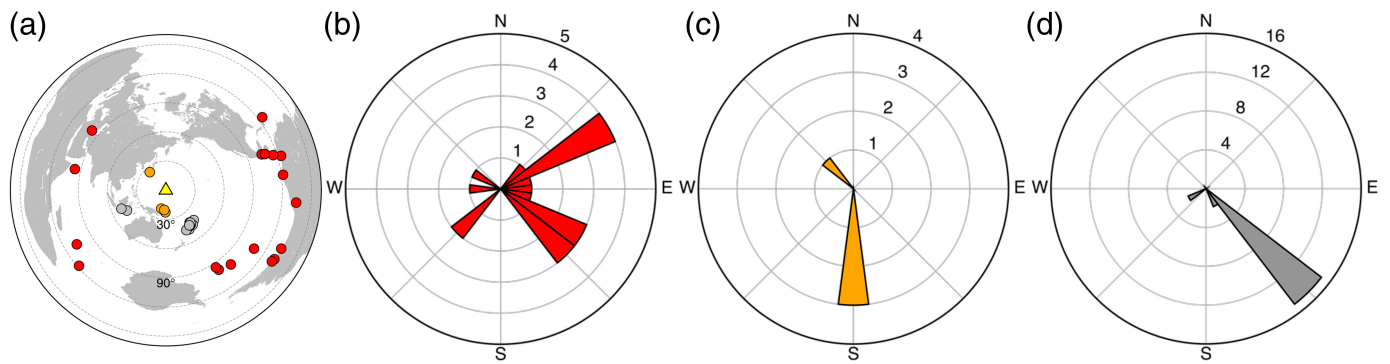
Data acquisition and preprocessing

The Oldest-1 array consisted of 12 BBOBSs deployed by free fall from October 2018 to November 2019 on the 160–180 Ma Pacific seafloor at water depths of 5600–6100 m. The spacing between adjacent stations ranged from 114 to 231 km (average spacing of 184 km), and each instrument was equipped with a CMG-3T broadband seismometer (Güralp Systems Ltd.) and a differential pressure gauge (DPG), recording at 100 samples per second. We used data from 11 BBOBSs, excluding station OL12, which recorded only DPG data and no three-component seismograms.

Before conducting SWS analysis, determination of sensor orientation was essential, particularly for ocean-bottom seismometers (OBSs), as free-fall deployment leads to unknown horizontal orientations. After removing instrumental responses, orientation angles of the horizontal components were examined following Stachnik *et al.* (2012), which determines station orientation using regional and teleseismic Rayleigh waves. The estimated orientation angles are summarized in Table S1, available in the supplemental material to this article, including the circular mean, median, median absolute deviation, and the number of events used in the analysis. The estimated orientation is consistent with a measurement observed by a magnetic compass installed in a BBOBS (see Data and Resources). The differences are less than about 30°, likely due to a magnetic field induced by the electric circuit of a BBOBS.

SWS analysis

We measured individual SWS using *XKS* phases (*SKS*, *SKKS*, and *SKIKS*) and *ScS* phases, as well as teleseismic direct *S* phases recorded by the 11 BBOBSs during the deployment period. For SWS analysis, teleseismic events with magnitudes



≥ 6.0 at epicentral distances between 85° and 140° were selected for XKS phases to avoid contamination from other phases and ensure high-quality waveforms. For ScS phases, events with magnitudes ≥ 5.5 at epicentral distances $< 30^\circ$ were used. Direct S phases were selected from events with magnitudes ≥ 5.3 and epicentral distances of 40° – 80° . Unlike XKS phases, which are influenced solely by receiver-side anisotropy, ScS and direct S phases can be affected by both source- and receiver-side anisotropy (Long and Silver, 2009). To reduce potential shallow source-side splitting effects, we restricted ScS and direct S events to those with focal depths greater than 350 and 200 km, respectively, following empirical selection criteria adopted in previous SWS studies (Long and van der Hilst, 2005; Kang and Shin, 2009). The distribution of selected events is shown in Figure 2, and the event selection criteria are summarized in Table 1.

SWS analysis was performed using the Pytheas software package (Spingos *et al.*, 2020), which employs three single-phase approaches to determine the FPD and delay time: the rotation-correlation (RC) method (Bowman and Ando, 1987), the minimum energy method (Silver and Chan, 1991), and the eigenvalue (EV) method (Silver and Chan, 1991). Among these, we primarily utilized the EV method, which performs a grid search for FPD and delay time by minimizing the second eigenvalue (λ_2) of the covariance matrix, a measure of the linearity of particle motion. The EV method, along with the RC method, is advantageous when the initial polarization direction of the shear wave is unknown (Silver and Chan, 1991; Long and Silver, 2009). Cluster analysis (Teanby *et al.*, 2004) was adopted to identify optimal splitting parameters from multiple time windows, thereby avoiding subjective window selection and ensuring robust estimates. Each window contains approximately one full cycle of the S phase after the arrival time to prevent cycle skipping. All data were band-pass filtered using a fourth-order Butterworth filter (0.05–0.125 Hz), a frequency band in which the probabilistic power spectral density of both horizontal components showed consistently low-noise levels (Kang *et al.*, 2023).

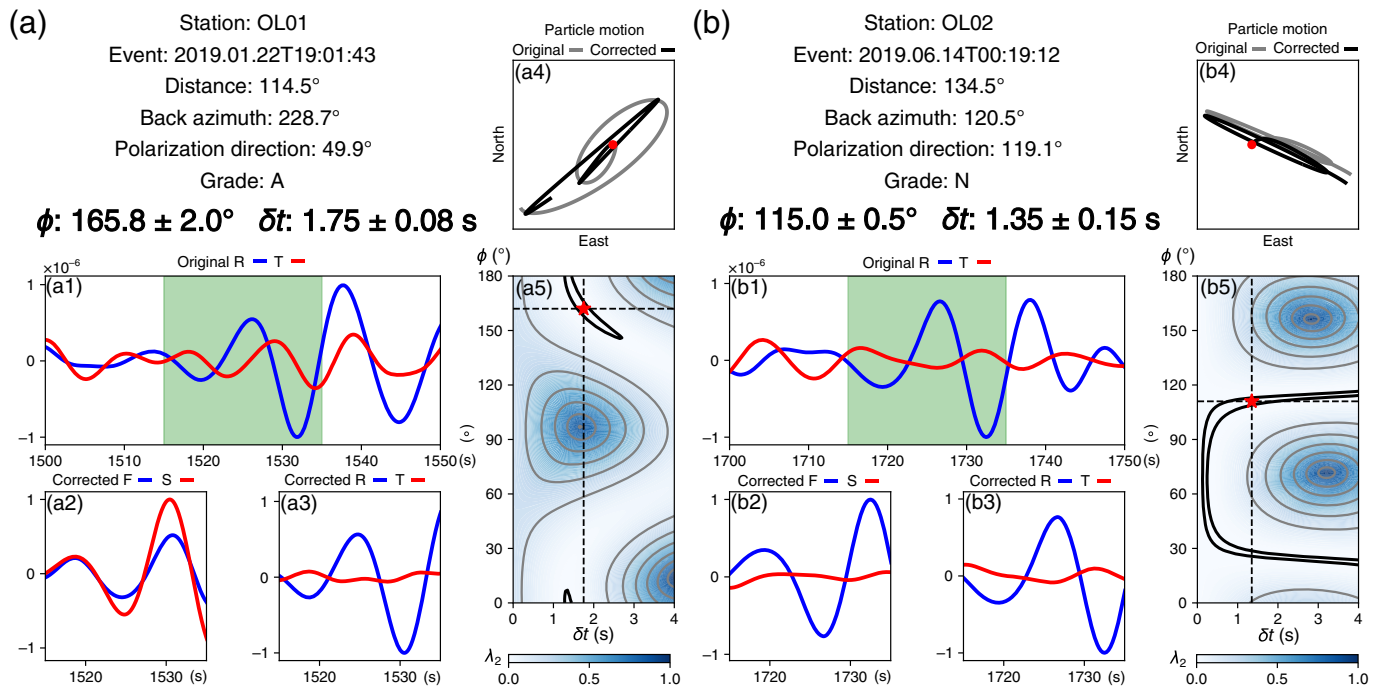
The measurements were filtered based on objective criteria adapted from Spingos *et al.* (2020) to ensure waveform quality and robust estimation of the FPD and delay time: (1) signal-to-

Figure 2. (a) Distribution of earthquakes that satisfied all selection criteria during the deployment period (October 2018 to November 2019). Red, orange, and gray filled circles represent XKS, ScS, and direct S events, respectively. The yellow triangle marks the location of the Oldest-1 array. (b) Rose diagram showing the azimuthal distribution of unique events used for shear-wave splitting (SWS) measurements across the Oldest-1 array (XKS; $N = 18$), with back azimuths calculated relative to station OL07 (near the array center). (c) Rose diagram showing the azimuthal distribution of unique events used for SWS measurements across the Oldest-1 array (ScS; $N = 4$), with back azimuths calculated relative to station OL07. (d) Rose diagram showing the azimuthal distribution of unique events used for SWS measurements across the Oldest-1 array (direct S; $N = 19$), with back azimuths calculated relative to station OL07. The color version of this figure is available only in the electronic edition.

noise ratio (SNR) > 3.0 ; (2) FPD and delay-time uncertainties $< 20^\circ$ and < 0.5 s, respectively; and (3) cross-correlation coefficients of the corrected waveforms (CC_{FS} and CC_{NE}) > 0.6 for the fast-slow and northeast components, respectively. SNR was defined as the ratio of the root-mean-square amplitude in the signal window (from the picked phase arrival to 5 s after) to that in the noise window (from 5 s before the phase arrival to the phase onset) on the horizontal components. Some of the measurements were excluded when the vertical-component amplitude exceeded that of the horizontal components at the phase onset, indicating severe noise contamination. Measurements failing any of these criteria were assigned grade “E” and excluded from further analysis.

TABLE 1
Parameters and Selection Criteria for Teleseismic Earthquakes Used in Shear-Wave Splitting (SWS) Analysis

Seismic Phase	XKS	ScS	Direct S
Distance ($^\circ$)	85–140	0–30	40–80
Magnitude (M_w)	≥ 6.0	≥ 5.5	≥ 5.3
Depth (km)	> 0	> 350	> 200
Number of events	18	4	19



For accepted measurements, a grade score (S) was calculated following (Spingos *et al.*, 2020):

$$S = \frac{\frac{\delta\phi}{\delta\phi_{\max}} + \frac{\delta t_d}{\delta t_{d,\max}} + \frac{1-|CC_{FS}|}{1-CC_{FS,\min}} + \frac{1-|CC_{NE}|}{1-CC_{NE,\min}}}{4}, \quad (1)$$

in which $\delta\phi$ and δt_d denote the uncertainties of FPD and delay time, respectively. $\delta\phi_{\max}$ and $\delta t_{d,\max}$ are their maximum allowable errors (20° and 0.5 s), and $CC_{FS,\min}$ and $CC_{NE,\min}$ represent the minimum acceptable correlation coefficients (0.6). Based on the calculated scores, data were classified into four grades: “A” (best; $S \leq 0.25$), “B” ($0.25 < S \leq 0.5$), “C” ($0.5 < S \leq 0.75$), and “D” (worst; $S > 0.75$). Only grades “A” and “B” were retained for subsequent analysis.

Grade “A” measurements show clear phase arrivals, distinct splitting in the uncorrected waveforms, and linear particle motion in the corrected horizontal components. Grade “B” measurements are less clear but still yield robust splitting parameters within acceptable error limits. Measurements were classified as null (grade “N”) when no resolvable splitting was observed, characterized by negligible transverse energy and linear particle motion before correction (Long and Silver, 2009). In this study, we quantified the linearity of the uncorrected particle motion using the two horizontal components, with values closer to 1 indicating more linear particle motion. Measurements were classified as nulls when their linearity exceeded 0.95 , and the difference between the initial shear-wave polarization, estimated from the uncorrected particle motion for null candidates, and the estimated FPD was within 15° of either parallel or perpendicular. Possible causes of null measurements include: (1) negligible azimuthal anisotropy along the ray path (Wüstefeld and Bokelmann, 2007; Long and Silver, 2009), (2) the presence of

Figure 3. Examples of individual SWS measurements for the XKS phase, illustrating (a) a grade A result and (b) a null (grade N) result. For each case, (a1–a4, b1–b4) show the uncorrected radial–transverse (R–T) components, the corrected fast–slow (FS) components, the corrected R–T components, and the particle motions before (gray) and after (black) correction. In (a2–a3) and (b2–b3), the traces are normalized by the maximum amplitudes of the two horizontal components for clarity, whereas (a1) and (b1) show the original, unscaled amplitudes. (a5) and (b5) display contour maps of the normalized second eigenvalue (λ_2) from the grid search used to determine the FPD (ϕ) and delay time (δt). The thick-black line marks the 95% confidence contour. The color version of this figure is available only in the electronic edition.

anisotropy, but with the initial polarization aligned subparallel or suborthogonal to the fast axis (Savage, 1999; Wüstefeld and Bokelmann, 2007; Long and Silver, 2009), or (3) cancellation of splitting in multilayered anisotropic structures (Barruol and Hoffmann, 1999; Saltzer *et al.*, 2000; Wüstefeld and Bokelmann, 2007).

The observed polarization directions were compared with theoretical directions predicted from focal mechanisms (for ScS and direct S phases) or event back azimuths (for XKS phases). Measurements with differences less than 30° from theoretical predictions were considered robust and retained. Figure 3 shows representative examples of grade A and N results. As a cross-check, the RC method was also applied to confirm consistency.

Results

A total of 128 SWS measurements (50 split and 78 null) were retained after applying quality-control criteria, including grade thresholds and polarization-direction checks (Table S2;

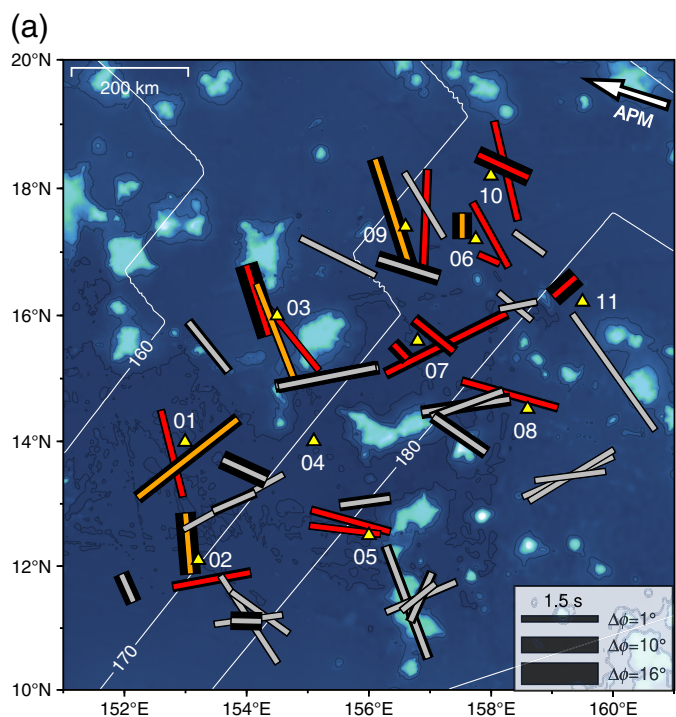


Fig. S1). Despite the inherently high noise levels of OBS recordings, high-quality SWS measurements were obtained after band-pass filtering at 0.05–0.125 Hz. Of the 128 measurements, 48 were from XKS (16 split and 32 null), 7 from ScS (5 split and 2 null), and 73 from direct S phases (29 split and 44 null). For the 50 splitting measurements, 80% of the dataset has FPD uncertainties below 9°, with the largest uncertainty reaching 16.8°. This narrow uncertainty distribution indicates that the estimated FPDs are robustly resolved.

Across the study region, the FPDs show considerable scatter but broadly define two dominant trends: east-northeast–west-southwest and northwest–southeast to north-northwest–south-southeast (Fig. 4a and Fig. S2). XKS measurements are mostly oriented northwest–southeast to north-northwest–south-southeast. The limited ScS measurements are compatible with this range, although one ScS measurement from OL01 shows a northeast–southwest orientation. Direct S phases additionally reveal northeast–southwest or east-northeast–west-southwest orientations. The average delay time for all nonnull measurements is 1.46 s (1.44 s for XKS, 1.69 s for ScS, and 1.43 s for direct S phases), which exceeds the global average of 1.0 s reported by Silver and Chan (1991).

Figure 4 shows individual SWS measurements plotted at ray-piercing points at a depth of 200 km. This depth was chosen as a common reference level to enable direct comparison with the *P*-wave tomography results (Kang et al., 2023) and to sample the asthenosphere upper mantle. For direct S phases, the 200 km ray-piercing points may be laterally offset from the station locations; therefore, interpretations of direct S measurements are guided by their behavior across multiple ray-piercing depths between 0 and 250 km, as shown in Figure S3.

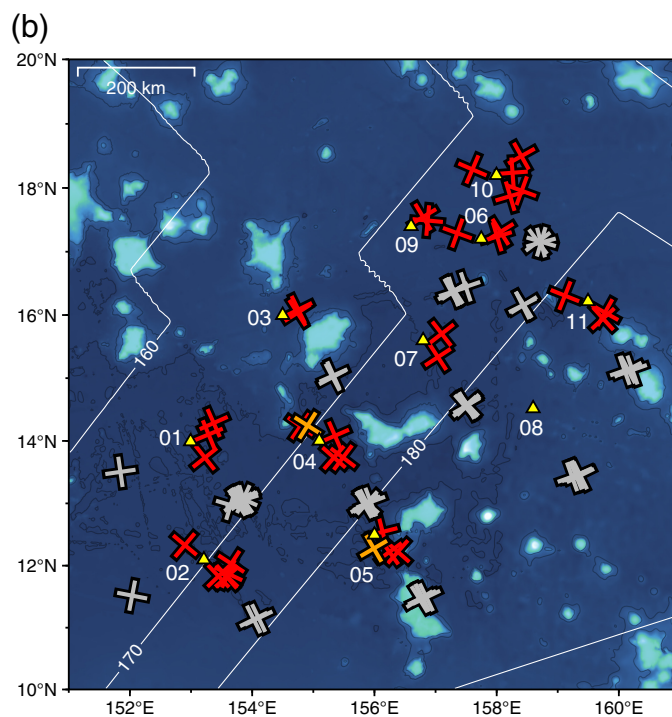


Figure 4. Individual SWS parameters plotted at the ray-piercing points at a depth of 200 km. Red, orange, and gray bars represent measurements from XKS, ScS, and direct S, respectively. (a) Nonnull splitting results. The orientation and length of each bar correspond to the FPD and delay time, respectively. Bar thickness represents the uncertainty in FPD, as shown in the legend. (b) Null measurements. The orientation of each cross symbol denotes the apparent fast or slow axis of anisotropy. Cross symbols are plotted with a fixed symbol size for visibility and not scaled by delay time. Other symbols are the same as in Figure 1. The color version of this figure is available only in the electronic edition.

FPDs across the array exhibit substantial variability, with each station showing distinct splitting behavior (Fig. 4a). At OL01, FPDs are inconsistent: the XKS phases arriving from the west yield north-northwest–south-southeast, whereas ScS arrivals from the east show northeast–southwest orientations. Direct S phases arriving from the east display relatively consistent northeast–southwest orientations (Fig. 4a and Fig. S3). At OL02, the ScS measurement from the west yields north-northwest–south-southeast orientations, whereas XKS measurements from the east exhibit east-northeast–west-southwest orientations. However, direct S phases show variable splitting behavior, even for nearly identical ray paths. At OL03, XKS and ScS phases display consistent north-northwest–south-southeast orientations, whereas the direct S phases entering from the southeast show east-northeast–west-southwest orientations. At OL04, only null measurements were obtained for XKS and ScS phases (Fig. 4b), whereas two direct S phases show east–west and west-northwest–east-southeast orientations (Fig. 4a and Fig. S3). At

OL05, XKS phases exhibit west-northwest–east-southeast orientations, whereas direct S phases are scattered but generally fall within northeast–southwest orientations.

In the northeastern region, the stations OL06, OL09, and OL10 show FPDs ranging from near east–west (west-northwest–east-southeast) to north-northwest–south-southeast, with a general pattern of north-northwest–south-southeast for measurements arriving from the north and near east–west orientations for others (Fig. 4a and Fig. S3). We observe two distinct groups of FPD at OL07: one shows northwest–southeast orientations (two XKS measurements), whereas the other displays east-northeast–west-southwest orientations (another XKS and two direct S phases with similar back azimuths). At OL08, a systematic contrast is evident, with direct S and XKS measurements arriving from the west showing northwest–southeast orientations, whereas those arriving from the southeast show east-northeast–west-southwest orientations. Finally, OL11 recorded predominantly null measurements (Fig. 4b); however, two nonnull measurements were also observed, yielding a northeast–southwest orientation with a small delay time (0.6 s) for an XKS phase and a northwest–southeast orientation with a large delay time (2.8 s) for a direct S phase (Fig. 4a and Fig. S3).

Discussion

Upper-mantle origin (> 50 km) of the observed seismic anisotropy

SWS measurements reflect the integrated effect of anisotropy along the ray path, motivating an assessment of the dominant depth contributions. XKS phases provide the most direct constraints on receiver-side anisotropy because their initial polarization is controlled by *P*-to-*S* conversion at the core–mantle boundary (CMB). In contrast, both ScS and direct S phases may include source-side contributions, although thresholds of focal depth were applied to reduce potential source-side effects. This limitation is particularly relevant because source-side splitting has been documented for earthquakes deeper than 200 or 350 km depth (e.g., Foley and Long, 2011; Mohiuddin et al., 2015; Roy et al., 2017). Most accepted direct S measurements in our dataset are from earthquakes deeper than 500 km, mainly in the Tonga–Fiji region where previous work using Tonga deep earthquakes suggested limited observable anisotropy near the deep earthquake foci and in the shallower lower mantle beneath the source region (Kaneshima, 2014). Nevertheless, this regional context does not rule out source-side contamination, particularly for ScS and direct S phases, and these measurements are therefore interpreted cautiously.

To further constrain the depth range relevant to the observed splitting, we consider whether shallower structures could contribute to a meaningful extent. Crustal anisotropy typically produces delay times no greater than 0.2 s (e.g., Crampin, 1994; Saiga et al., 2003), except in regions with

exceptionally thick crust such as the Tibetan Plateau (~80 km), where values up to 0.5 s have been documented (e.g., Zheng et al., 2018). A prior geophysical survey indicates that crustal thickness beneath the study area is <8 km (Abrams et al., 1993), implying that crustal contributions are negligible relative to the observed delay times (average of 1.46 s).

Although most SWS investigations attribute splitting to upper-mantle anisotropy, only a few studies have identified anisotropy in the lowermost mantle (e.g., Vinnik et al., 1989; Niu and Perez, 2004; Lutz et al., 2020). Such anisotropy is predominantly confined to the margins of the large low-shear-velocity provinces (LLSVPs) beneath the Pacific and Africa (e.g., Lynner and Long, 2014a; Reiss et al., 2019). In contrast, the ray paths of XKS and ScS phases mainly traverse the interior of the Pacific LLSVP, a region generally regarded as isotropic just above the CMB (Romanowicz and Wenk, 2017). If lower-mantle anisotropy were the dominant source, ray paths sampling similar regions of the lowermost mantle would be expected to yield broadly consistent splitting parameters. However, the observed FPDs vary among these measurements despite their comparable lowermost mantle sampling. Specifically, several XKS and ScS measurements arriving from the northwest sample similar piercing points at ~2800 km depth, yet exhibit contrasting FPDs. Although only a few XKS and ScS cases allow this comparison, the available observations do not support a coherent anisotropic layer at the base of the mantle. Therefore, we infer that upper-mantle anisotropy is the dominant source of the observed splitting.

To place additional constraints on the depth of anisotropy, we compare our results with surface-wave observations. Kawano et al. (2023) investigated azimuthal anisotropy using surface-wave dispersion from the Oldest-1 array and reported average fast-axis directions to depths of 50 km, with northwest–southeast orientations in the west and northeast–southwest in the east (Fig. 1). In contrast, our splitting results reveal notably different FPDs than those inferred from surface-wave dispersion (Fig. 4a). In the western region, most FPDs trend more northward than those of Kawano et al. (2023), although two XKS measurements at station OL05 align with their northwest–southeast pattern. In the eastern region, our measurements generally exhibit northwest–southeast or north-northwest–south-southeast orientations, which differ markedly from the northeast–southwest directions reported by Kawano et al. (2023), although a limited number of measurements at stations OL07, OL08, and OL11 show northeast–southwest directions. Such discrepancies between anisotropy inferred from surface-wave dispersion and SWS observations have also been reported in other regions and are commonly attributed to differences in depth sensitivity or vertically varying anisotropic structure (e.g., Wüstefeld et al., 2009; Pandey et al., 2015).

To assess the potential contribution from the uppermost mantle, we estimate the splitting delay time expected from the azimuthal anisotropy reported by Kawano et al. (2023).

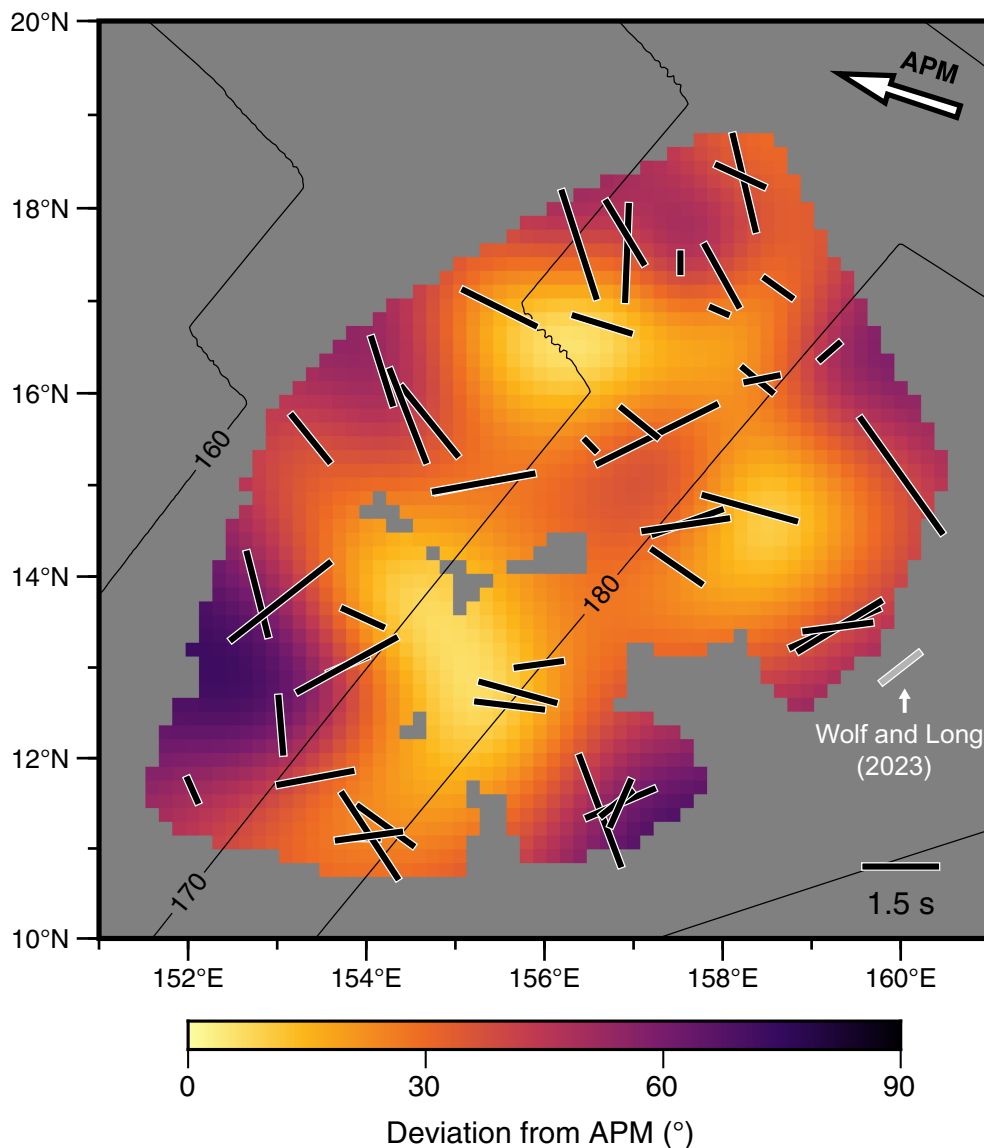


Figure 5. Map of interpolated deviations from the APM. Black bars represent SWS measurements with their orientation and length corresponding to the FPD and delay time, respectively. The gray bar indicates a PS-phase splitting measurement from Wolf and Long (2023). Other symbols are the same as in Figure 1. The color version of this figure is available only in the electronic edition.

For this first-order estimate, anisotropy is assumed to be uniform within the study region. The expected delay time can be approximated using a simple analytical relation:

$$\delta t \approx \frac{L \times A}{V_s}, \quad (2)$$

in which L denotes the thickness of the anisotropic layer, A is the magnitude of azimuthal anisotropy, and V_s is the shear-wave velocity. Kawano *et al.* (2023) reported average peak-to-peak amplitudes of azimuthal anisotropy of $2.8 \pm 1.0\%$ and $1.6 \pm 0.9\%$ for the western and eastern parts of the study area, respectively, for depths between the Moho and 50 km. Following

Abrams *et al.* (1993), Moho depths of 7.9 km in the west and 7.6 km in the east were adopted in that study. Over this depth range, an average isotropic shear-wave velocity of approximately 4.7 km/s was inferred. Using these values, the maximum splitting delay times expected from Moho to 50 km depth are approximately 0.25 ± 0.09 s in the western region and 0.14 ± 0.08 s in the eastern region. These values are substantially smaller than the observed average delay time of ~ 1.5 s, indicating that anisotropy confined to the uppermost 50 km contributes only a minor fraction of the measured splitting. In this context, the systematic mismatch between fast-axis directions inferred from surface-wave dispersion and FPDs from SWS measurements is most naturally explained by differences in depth sensitivity. This suggests that the dominant contribution to the observed splitting arises from anisotropy at depths greater than ~ 50 km.

Regional variability of FPDs across the Oldest-1 array

The obtained SWS parameters exhibit substantial variability depending on station location, back azimuth, initial S-wave

polarization, and ray-path geometry, including incidence angle of incoming phases. Although several measurements sampling the central portion of the array show FPDs broadly consistent with the N73°W APM direction, the FPDs near the margins display markedly different orientations (Fig. 5).

In the northern and western parts of the region, FPDs tend to differ from the APM (by $\sim 70^\circ$) and from the approximately N55°W paleospreading direction inferred from the regional fracture-zone trend of 160–180 Ma seafloor (which is nearly perpendicular to age contours; Abrams *et al.*, 1992; Nakanishi *et al.*, 1992). In particular, splitting measurements at the northern stations (OL03, OL06, OL09, and OL10) show a tendency toward more northward orientations relative to the

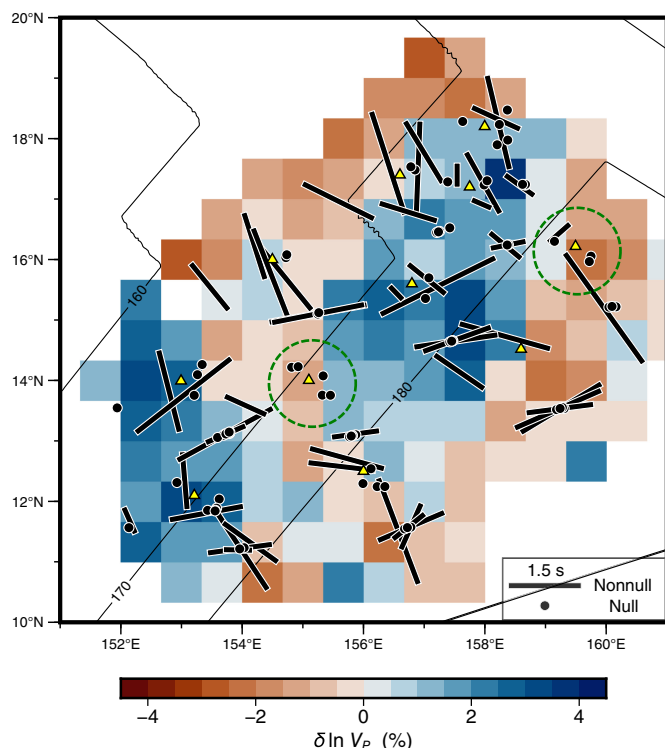


Figure 6. Individual SWS measurements superimposed on *P*-wave velocity anomalies at a depth of 185 km (Kang *et al.*, 2023). Black bars indicate non-null splitting measurements, where bar orientation and length represent the FPD and delay time, respectively. Black circles denote null measurements. Green dashed circles highlight the vicinity of stations OL04 and OL11, showing a high proportion of null results relative to non-null measurements. Other symbols are the same as in Figure 1. The color version of this figure is available only in the electronic edition.

APM, with angular deviations exceeding the estimated uncertainties of individual measurements. At OL03, several FPDs are subparallel to the fracture-zone-based spreading direction. Although this geometric correspondence does not uniquely constrain the depth of anisotropy, it provides an alternative structural reference distinct from the present-day APM. These regional patterns suggest that simple APM-parallel deformation alone cannot fully account for the observed anisotropy (Fig. 5 and Fig. S4). In the southeastern area, two clusters of SWS measurements sampling nearly identical ray paths show robust northeast–southwest orientations, deviating from the present-day plate motion direction. A nearby teleseismic *PS* splitting measurement reported by Wolf and Long (2023) exhibits a nearly identical FPD (Fig. 5), providing independent support for the reliability of these observations.

Although direct *S* phases are more sensitive to incidence-angle effects than *XKS* and *ScS* phases and may therefore carry additional interpretational uncertainty (e.g., Song and Kawakatsu, 2012; Appini *et al.*, 2025), the overall pattern of FPD variability described earlier does not depend on a single

phase type. These observations suggest that factors beyond simple plate-motion-related shear are required to explain the observed spatial variability in FPDs.

Constraints on mantle flow around low-velocity anomalies and a high-velocity lithospheric residue

We compared our SWS measurements with *P*-wave tomography results derived from the same Oldest-1 stations (Kang *et al.*, 2023) to examine how mantle flow interacts with lateral velocity heterogeneities beneath the Pacific Triangle (Fig. 6). A striking feature is the predominance of null measurements at stations OL04 and OL11 (Fig. 6, green dashed circles), where pronounced low-velocity anomalies persist throughout the imaged depth range (35–305 km). As noted earlier, null results generally arise when seismic rays do not traverse azimuthally anisotropic structures, or when the initial polarization is nearly parallel or orthogonal to the fast axis. At OL04, all five *XKS* and *ScS* measurements are null, even though they arrive from different back azimuths. These measurements do not indicate a consistent fast-axis orientation, suggesting that the null results cannot be explained by the initial polarization being coincidentally parallel or orthogonal to the fast axis. Instead, the repeated occurrence of nulls across different ray paths suggests weak or absent azimuthal anisotropy beneath OL04. Considering the strong low-velocity anomalies and the station's proximity to the MST, an alternative interpretation is localized mantle upwelling, which can generate dominantly vertical deformation fabrics and thus produce minimal azimuthal anisotropy.

Similarly, OL11 is dominated by nulls, with only one splitting measurement among *XKS* phases and one among the direct *S* phases (Fig. 4b). Nulls are observed over multiple back azimuths, although the small number of observations prevents a definitive interpretation. The single-splitting measurement for the *XKS* phases shows an FPD of 49° with a small delay time (0.6 s), distinct from neighboring measurements, supporting the presence of weak azimuthal anisotropy beneath OL11. Given the nearby low-velocity anomaly, this pattern may also reflect localized mantle upwelling as proposed for OL04.

In contrast to these low-velocity regions, our SWS results show a broad spatial organization of FPDs in the vicinity of a high-velocity anomaly (HVA) located at 155°–159° E and 14°–18° N (Figs. 6 and 7). The spatial averaging shown in Figure 7 is intended to smooth local variability in individual SWS measurements and to emphasize regional trends in FPDs across the study area (Text S1). To minimize interpretational complications associated with *ScS* and obliquely incident direct *S* phases, the averaging shown here is based on *XKS* measurements only; corresponding results including all phase types are provided in Figure S5 for comparison. Although local differences appear in the western part of the array, where *XKS* sampling is sparse and measurements from *ScS* and direct *S* phases show different orientations, the first-order FPD pattern around the central to eastern HVA is retained in the *XKS*-only

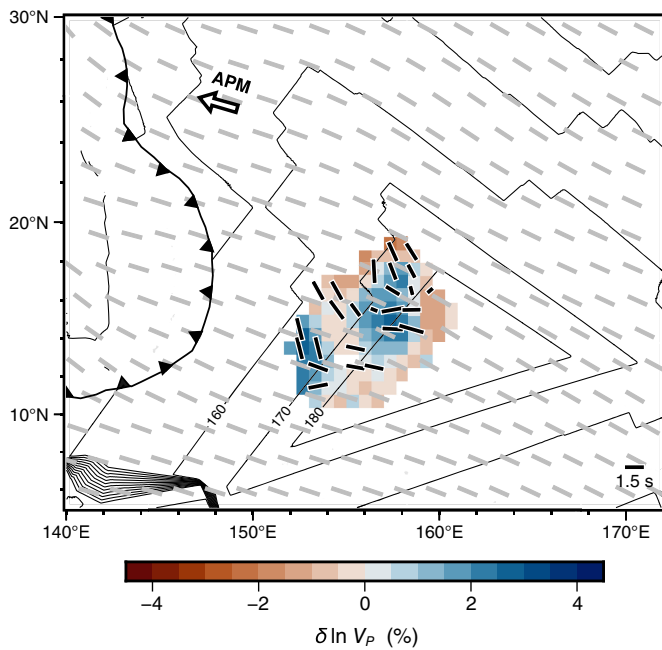


Figure 7. Spatial distributions of SWS measurements from XKS phases projected onto ray-piercing points at a depth of 200 km (black bars), overlain on P -wave velocity anomalies from Kang *et al.* (2023). Measurements are averaged within $1^\circ \times 1^\circ$ grid cells to highlight regional trends. Gray lines in the background indicate lattice-preferred orientation (LPO) predictions at 200 km depth from Becker *et al.* (2014). Other symbols are the same as in Figure 1. The color version of this figure is available only in the electronic edition.

average. This HVA, ~ 200 km in radius, extends from 95 to 185 km depth and is 2%–4% faster than the surrounding mantle. Because the tomographic model of Kang *et al.* (2023) was obtained from an isotropic inversion of teleseismic P -wave travel-time residuals, unmodeled seismic anisotropy may contribute to the amplitude or geometry of the imaged anomaly, as also discussed by Kang *et al.* (2023). With this caveat in mind, Kang *et al.* (2023) interpreted this fast feature, in combination with 1D conductivity models derived from the OBEM data reported by Kim *et al.* (2020), as a cold, dry, and rheologically strong lithospheric body, potentially representing a remnant of melt-depleted lithosphere that has persisted within the upper mantle.

The FPDs surrounding the HVA exhibit systematic variations in orientation, with west-northwest–east-southeast orientations to the south and north-northwest–south-southeast orientations to the west and north (Fig. 7). Although this spatial relationship is nonunique, such patterns are consistent with a range of possible interactions between mantle deformation and mechanically strong lithosphere, including perturbation or redirection of asthenospheric flow around a rigid lithospheric body. Similar anisotropic patterns have been reported around regions of thick lithosphere in various tectonic settings,

including eastern North America (Fouch *et al.*, 2000), the Ordos block of the North China craton (Li *et al.*, 2020), the Songliao basin in northeast China (Xu *et al.*, 2023), and the Moesian lithospheric block in central and eastern Europe (Petrescu *et al.*, 2020). A comparable feature is observed around the Ontong Java Plateau (OJP), located ~ 1500 km south of our study region, where systematically deflected FPDs along the plateau margins have been attributed to mantle flow diverted around a thick lithospheric root (Klosko *et al.*, 2001). Isse *et al.* (2021) interpreted the thickened OJP lithosphere as dry, depleted residual material underplated beneath pre-existing oceanic lithosphere during thermochemical plume activity, allowing rheologically strong material to remain attached to the Pacific lithosphere over geologic time. Although the origin of the HVA beneath the Pacific Triangle cannot be uniquely determined, potential tectonic links between western Pacific basin magmatism and the OJP system (Castillo *et al.*, 1994) suggest that the OJP provides a useful analog for considering how inherited thermochemical lithosphere or melt-depleted residue could form a mechanically strong lithospheric domain.

In this context, the HVA imaged beneath the Pacific Triangle, although potentially affected by unmodeled anisotropy, provides a structural context for interpreting the regional-scale FPD variations. Under this scenario, a long-lived, mechanically strong lithospheric remnant may locally perturb APM-parallel asthenosphere flow, potentially including flow deflection around the anomaly. Consistent with this interpretation, delay times near the center of the HVA (0.99 s) are smaller than those in the surrounding region (1.55 s), aside from one outlier at station OL07. This reduction may reflect a diminished contribution from anisotropic asthenosphere mantle along ray paths traversing the thick lithospheric remnant, resulting in weaker path-integrated splitting relative to adjacent regions.

Depth-dependent and structurally complex anisotropy beneath the Pacific triangle

Several observations, particularly from southern stations and from within the HVA, indicate that the observed splitting patterns cannot be explained by a simple, single-layer structure. Direct S -phase measurements at OL02 and OL05 in the south show variable FPDs (north-northwest–south-southeast to northeast–southwest; Fig. 4a), even for similar receiver-side ray paths within the upper mantle. Because direct S phases may include source-side contributions, we do not use these measurements alone to infer receiver-side complexity. Instead, a clearer receiver-side constraint comes from XKS phases at OL07, where two measurements from the northeast and southwest yield northwest–southeast orientations with small delay times (0.4 and 1.0 s) whereas one measurement from the southeast shows a northeast–southwest orientation with a much larger delay time (2.6 s).

Differences in SWS parameters between XKS phases and direct S phases at most stations further indicate that the

two phase types do not yield a uniform single-station splitting pattern. However, these differences should not be attributed solely to receiver-side anisotropic complexity due to the possible source-side contributions to direct S phases. Therefore, we use the phase-dependent differences mainly as a diagnostic of structural complexity. Stacked measurements of XKS, direct S (DIR), and combined (ALL) phases (Text S2; Figs. S6 and S7) support this interpretation: except for northern stations (OL03, OL06, and OL10), most stacks exhibit unstable or inconsistent results, with FPD differences exceeding 30° between XKS and DIR cases. Moreover, the ALL stacks do not substantially reduce the uncertainties at many stations, suggesting that the observed splitting patterns cannot be adequately represented by a simple single-layer anisotropic structure.

Taken together, these observations are more consistent with depth-dependent and/or laterally heterogeneous anisotropy beneath the oldest Pacific plate, rather than a simple single-layer structure. At the same time, it is also worth noting that apparent SWS measurements can be sensitive to incidence angle and the geometry of anisotropic structures, which may contribute to variability in the observed splitting results (Song and Kawakatsu, 2012, 2013). Although forward modeling of layered anisotropy can aid interpretation of complex SWS observations (e.g., Silver and Savage, 1994), the limited data coverage and heterogeneous ray-path sampling in our study, in which individual measurements sample different mantle volumes, preclude meaningful constraints from such tests. Therefore, we restrict our discussion to observational evidence for depth-dependent anisotropy.

Surface-wave studies of the Pacific have shown vertically stratified anisotropy: at shallow depths of ~100 km, FPDs tend to align with fossil spreading directions, whereas at greater depths they are more closely parallel to present-day plate motions (e.g., Savage, 1999; Smith *et al.*, 2004). This depth-dependent pattern is commonly interpreted as fossil lithospheric anisotropy overprinted by present-day asthenosphere flow. More recent regional OBS studies in the Pacific indicate that azimuthal anisotropy within the lithosphere–asthenosphere system can vary with depth and need not follow a simple relationship with either ancient spreading directions or present-day plate motion (Lin *et al.*, 2016; Takeo *et al.*, 2018). Tectonic reconstructions further reveal that the Pacific plate has migrated along multiple directions (east-northeast–west-southwest, north-northwest–south-southeast, and west-northwest–east-southeast) through time (Boschman and van Hinsbergen, 2016), and has passed over several hotspots, including Samoa, Rarotonga, and Rurutu-Arago (Wessel and Kroenke, 2008; Jackson *et al.*, 2024). Such hotspot interactions may have modified or partially reoriented pre-existing lithospheric fabrics, contributing to the complex anisotropy observed here.

Kawano *et al.* (2023) proposed a rejuvenation process in the study region linked to Cretaceous volcanic activities (~70 to

100 Ma), potentially reducing the influence of ~180 Ma fossil anisotropy. However, such rejuvenation might not have significantly modified the deepest portions of the lithosphere if a relatively thick lithospheric remnant limited melt infiltration and deformation at depth. In that case, some older fabrics could plausibly have been retained. The absence of prominent seamounts within the HVA is also consistent with the presence of thick lithosphere. Jackson *et al.* (2024) argued that Cretaceous plume-related volcanism failed to penetrate the exceptionally thick lithosphere beneath the OJP. A similar mechanism may apply beneath the HVA: if rejuvenation-related melts were unable to breach a thick lithospheric block, the deeper lithosphere could have escaped modification and preserved older fabrics. If multiple generations of fossil fabrics with differing orientations are vertically stacked, their contributions to SWS measurements may interfere destructively. This could produce smaller apparent delay times near the central HVA than in the surrounding regions, consistent with the observed spatial pattern of delay-time variations.

Although the observations discussed above point to vertically stratified anisotropy, certain regional patterns are not readily explained by depth-dependent fabrics alone. In particular, relatively consistent north-northwest–south-southeast FPDs are observed at northern stations (OL03, OL06, and OL10), and stations in the western part of the array (OL01 and OL02) exhibit systematic back-azimuth-dependent variations. At these western stations, arrivals from the west yield north-northwest–south-southeast FPDs, whereas arrivals from the east show more variable orientations. The relative consistency of northward-trending FPDs at the northern stations is not explained by depth-dependent anisotropic fabrics alone, suggesting that additional factors may contribute to the observed regional patterns. Alternative interpretations can therefore be considered. For example, the relatively uniform north-northwest–south-southeast FPDs at northern stations can plausibly reflect deflected asthenosphere flow around thick lithosphere, whereas the back-azimuth-dependent FPDs observed at western stations could arise from vertically layered anisotropy or laterally heterogeneous fabric rather than from laterally distinct mantle flow domains. Another possibility is that these patterns could indicate contributions from regional-scale mantle flow components in the northern and western parts of the study area that are not aligned with the present-day APM.

Although SSC has been discussed as a possible process beneath old oceanic lithosphere (e.g., Lynner and Long, 2014b; Kang *et al.*, 2023; Kawano *et al.*, 2023), our observations do not provide clear evidence for active SSC beneath the study area (Figs. 6 and 7). The relatively large average delay time (~1.5 s) exceeds those typically expected from SSC-induced flow, which would produce weaker azimuthal anisotropy (van Hunen and Čadež, 2009). Rather than arising from localized convective instabilities, the observed anisotropic patterns are more consistent with laterally coherent mantle

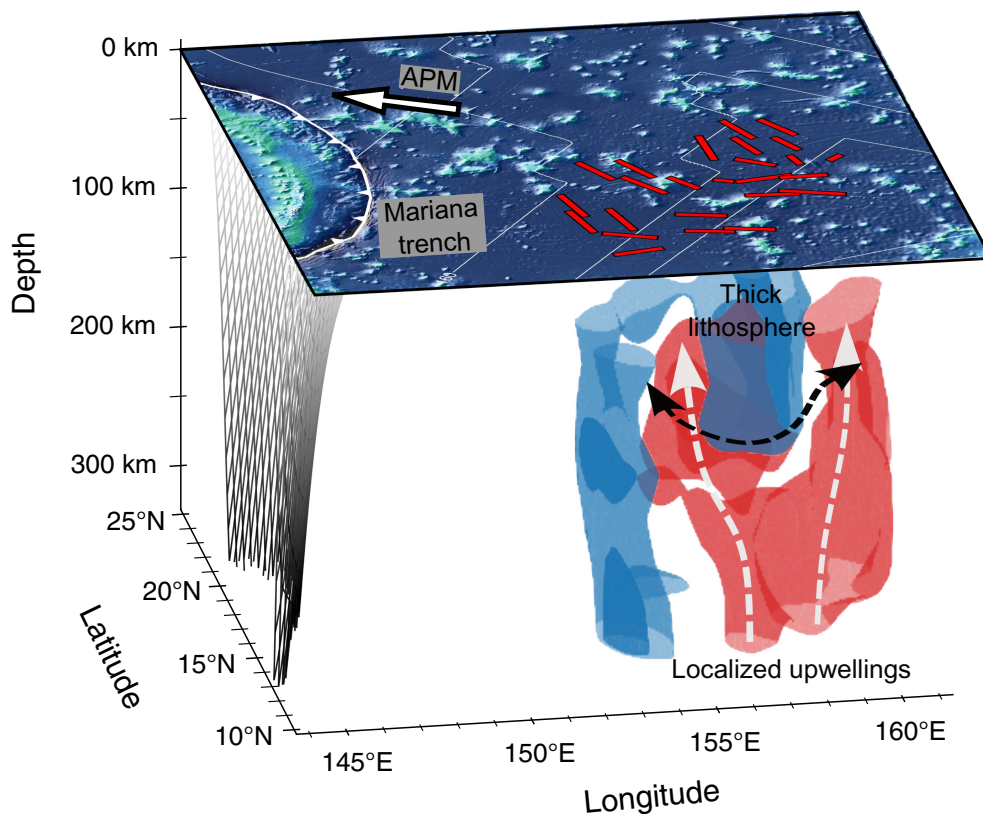


Figure 8. Schematic illustration of upper-mantle structure and inferred mantle flow beneath the Oldest-1 array in the Pacific Triangle. Red bars represent spatially averaged SWS measurements, shown in the same manner as the gridded orientations in Figure 7, but color coded for schematic clarity. Blue and red volumes show smoothed seismic velocity anomalies between ~100 and 300 km depth from *P*-wave tomography (Kang *et al.*, 2023). The blue volumes are based on positive velocity anomalies exceeding +1.0%, whereas the red volume is based on negative velocity anomalies below −0.9%. The subducting Pacific slab is shown to provide tectonic context. A dashed-black arrow indicates inferred horizontal mantle flow interacting with thick lithosphere, whereas dashed-white arrows indicate schematic localized upwellings. The color version of this figure is available only in the electronic edition.

deformation, which may include modification of flow around a lithospheric remnant in the east, and a systematic north-northwest–south-southeast orientation of FPDs in the northern and western parts of the study area (Fig. 8).

Conclusions

SWS measurements from the Oldest-1 OBS array provide new constraints on upper-mantle seismic anisotropy beneath the oldest preserved part of the Pacific plate. The observations reveal two dominant FPDs and relatively large delay times, indicating that the anisotropic structure cannot be explained by a simple single-layer model. The central to eastern part of the study region shows systematic variations in FPDs around a HVA that may represent a thick lithospheric body, providing a possible structural context for interaction between mantle deformation and mechanically strong lithosphere. Within this region, the reduced delay times and back-azimuthal dependence of splitting

parameters point to vertically superposed and structurally complex anisotropy. In the northern and western parts of the study region, FPDs cluster between north-northwest–south-southeast and north-west–south-east and remain distinct from the APM, even when measurement uncertainties are considered. This indicates that mantle flow beneath this part of the Pacific plate is not fully captured by simple plate-motion-driven mantle deformation, and may involve additional regional-scale mantle flow processes.

Taken together, these results indicate spatially heterogeneous upper-mantle deformation beneath the Pacific Triangle, shaped by interactions with thick, mechanically strong lithospheric structures and by depth-dependent anisotropy within the lithosphere and asthenosphere. Although the path-integrated nature of SWS measurements requires caution in assigning all variability to receiver-side structure, the XKS-based patterns and the consistency of the broader spatial trends provide

first-order constraints on mantle deformation beneath the old Pacific plate.

Data and Resources

Seismic waveform data used in this study were acquired during the Oldest-1 broadband ocean-bottom seismometer (BBOBS) experiment. The BBOBS dataset is openly available from the Ocean Hemisphere Project Data Management Center (OHPDMC; OHPDMC, 2023; <http://ohpdmc.eri.u-tokyo.ac.jp/dataset/campaign/obs/index.html>) following the end of the moratorium period in September 2023. Shear-wave splitting (SWS) analyses were conducted using the Pytheas software package (Spingos *et al.*, 2020). The corresponding individual SWS results are provided in the supplemental material. Figures were produced using PyGMT version 0.15.0 (<https://www.pygmt.org/v0.15.0>; Uieda *et al.*, 2021), a Python library based on Generic Mapping Tools (GMT) version 6 (<https://www.generic-mapping-tools.org>; Wessel *et al.*, 2019), as well as using the Matplotlib version 3.8.4 (<https://matplotlib.org/3.8.4>; Hunter, 2007) and Inkscape version

1.4.2 (<https://inkscape.org/release/inkscape-1.4.2>). P-wave tomography results shown in Figures 6–8 were taken from Kang *et al.* (2023). The lattice-preferred orientation (LPO) model used for comparison in Figure 7 is from Becker *et al.* (2014). Absolute plate motion (APM) is based on the model of Beavan *et al.* (2002). The OLDEST1 station map and list are available at <http://ohpdm.eri.u-tokyo.ac.jp/dataset/campaign/obs/oldest1/index.html>. All websites were last accessed in September 2026. The supplemental material for this article includes text, two tables, and seven figures.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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